

CHRISTOPHER HUANG

Fremont, CA | chrhua802@gmail.com
High School GPA: 3.88/4 (unweighted), 4.33/5 (weighted)

EDUCATION

Irvington High School (2023-Present)

Fremont, CA

AP Courses (9): AP English Language and Composition, AP Physics C, AP Physics 1, AP Statistics, AP Calculus AB, AP US History, AP World History, AP U.S. Government and Politics, AP Chinese

Awards and Honors: AP Scholar with Distinction Award (2026), Eagle Scout (2026), Northern/Central California Invention Convention Finalist (2026), President's Volunteer Service Award - Gold (2025, 2024, 2023), Conrad Challenge Innovator (2025), Rotary Young Innovators 3D Printing Challenge finalist (2026, 2025), Sparkhub Foundation Personal Achievements Award (individual) & Best Initiative Award (team) (2026)

LEADERSHIP ROLES

Big Hands e-NABLE

Fremont, CA

Founder/President

Dec. 2025-

Present

- Founded and led a chapter of the global e-NABLE the Future organization to develop free 3D-printed prostheses for the limb-different community.
- Coordinated with the World Action Fund and donated 15 custom prostheses to recipients in the West Bank and Uganda. Currently developing 40+ Ugandan cases.
- Led the development and dissemination of amputation specific finger prosthetics, transhumeral prosthetics, and cosmetic covers for the e-NABLE community.
- Raised over \$4k through community events, sports tournaments, and local donor outreach to support materials, production, and delivery.
- Mentored the Big Hands Next-Gen team, now grown to 30+ members. Taught CAD, 3D printing, and prepared junior members for technical and community driven contributions to e-NABLE.

Irvington High School Maker Club

Fremont, CA

Executive Officer/Publicity Officer/Secretary

Nov. 2024-Present

- Expanded club traction within the school through Instagram posts, meeting documentation, member organization, and event publicity, contributing to club growth from 40 to 130+ members.
- Founded the Prosthetics Branch of the club, teaching students general CAD design skills, prosthetic actuation technologies, and connecting students with design opportunities in the e-NABLE community.
- Actively participated in the Maker Branch through projects and external events like Invention Convention.
- Organized the first SpeedCAD and CAD Design competitions in the school.
- Contributed to general club management work and upkeep.

Scouting America Troop 488 (a FCSN affiliated troop for scouts with special needs)

Bay Area, CA

SPL/ASPL

Aug. 2024-Present

- Taught scouting skills in a designed setting to better accommodate scouts with special needs.

- Helped new SPL's adapt activities and general Scouting protocols for scouts with special needs.
- Planned, organized, and led interactive and specially designed troop meetings and outings.

Scouting America Troop 468

Bay Area, CA

Eagle Rank/ASPL/various other leadership positions

May 2020-Present

- Consistently attended 200+ monthly meetings, outings, or community service events across 6 years.
- Planned and led troop activities including meetings, outings, and community service projects.
- Developed revisions for the troop's requirement system to facilitate more involved teaching practices.
- Addressed dwindling participation amongst senior scout leadership by hosting sessions to promote confident speaking and mentorship to younger scouts during outings and meetings.
- Mentored junior scouts by teaching core Scouting skills and encouraged advancement through ranks.
- Led and completed my Eagle project that landscaped and constructed habitat planter beds for Math Science Nucleus at Tule Ponds at Tyson Lagoon to bolster local pollinator populations.

Camp Emerald Bay

Catalina Island, CA

Youth Staff

Summer of 2024

- 4-week staffing position in the Scouting summer camp to perform leadership training and practices

ACADEMIC EXPERIENCES

SDSU Yang Lab Research

San Diego, CA

Student Researcher

Jun. 2026-Sept. 2026

- Developed a metamaterial 3D printed anthropomorphic hand, focusing on sensor integration, object detection, and AM-based functional prototyping; achieved 98.1% accuracy in object detection using random forest ML.
- First author of a research manuscript entitled "*Additive Manufacturing of an Anthropomorphic Hand using Force Resistive Sensors for Object Detection*," which has been submitted to the Journal of Future Science and Engineering (JFSE) for publication.

Independent Research

Fremont, CA

Student Researcher

Jan. 2026-present

- To optimize remote prosthetic sizing, derived a Cauchy-based silhouette measurement algorithm; eliminated COLMAP's dense reconstruction overhead; validated 97.5–99.8% spectral accuracy via Fourier analysis across 33 OpenLimbTT models and benchmarked against VGGT to identify failure modes.
- Authoring a manuscript entitled "*Measure Before You Reconstruct: Remote Measurement of Residual-Limb Circumference from Sparse Smartphone Silhouettes*."

Student Researcher

Sept. 2025-Jul. 2026

- Researched applications of AI in additive manufacturing design and geometry optimization. Authored a manuscript entitled "*Artificial Intelligence-Driven Design and Geometry Optimization in Additive Manufacturing*," which has been submitted to the National High School Journal of Science (NHSJS).

UCSB Summer Research Academies (SRA)

Santa Barbara, CA

SRA Researcher

Jun. 2025-Jul. 2025

- Conducted 4-week research in Track 12 "Digital Brains."
- Researched how variation in autism diagnostic traits influences susceptibility to radicalization.
- First author of a research paper "*Evaluating Radicalization through Autism Diagnosis Criteria and Media Addiction*."

Fremont Student Makers (FSM)

Fremont, CA

CubeSat Structures Team

Sept. 2024-Present

- Trained under Irvington High School's GIS CubeSat team during the creation of Prometheus Alpha, now launched into space.
- Contributed to designing and editing the structure, as well as organizing machining protocols for space-grade parts of the [Maker^3 1U CubeSat](#), projected to launch in 2027.

Conrad Challenge 2025

Product Designer and Video Maker

Fremont, CA

Aug. 2024-Jan. 2025

- Invented and developed a dew collector, the major innovation of our solar panel self-cleaning system [SoliC](#).
- Contributed to the adoption of technologies including super-hydrophobic and super-hydrophilic coatings on solar panels.
- Recognized as a Conrad Challenge 2024-2025 Innovator.

Rotary Young Innovators 3D Printing Challenge

Finalist (2x)

Fremont, CA

Apr. 2025-Apr. 2026

- Designed a 3D-printed shoe sole embedded with piezoelectric sensors to harness kinetic energy generated by walking or running (2025).
- Designed a load bearing upper body exoskeleton to support safe manual handling and prevent injury in high-strain environments (2026).

XCamp Academy

Student

Bay Area, CA

2022-2023

- Took XCamp's programming courses from #102 (Python) to #300 (C++)

OTHER EXPERIENCES

Friends of Children with Special Needs (FCSN)

Volunteer

Bay Area, CA

Jan. 2023-Present

- Assisted children with special needs through interactive activities such as 1:1 tutoring and afterschool classes.

Irvington High School Color Guard

Color Guard

Fremont, CA

Aug. 2023-Dec.2024

- Attended regular rehearsals throughout the Fall semester and summer camps.
- Participated in football game performances and local and regional competitions.

Mini Cat Town

Volunteer

Milpitas, CA

Jun. 2023-Jul.

2024

- Supported day-to-day care of the kittens on the adaption floors in the Great Mall location.
- Interacted with guests and adopters.

White Stag

Camper in Phases 1,2, and 3

Monterey/High Sierra, CA

Summer of 2021, 2022, 2023

- White Stag member in Phases 1, 2, and 3 during the summer to participate in leadership training and practices.

SKILLS/INTERESTS

- Computer Software: Intermediate in Python/C++/Java/ HTML/CSS; Microsoft Office and Google Suite
- 3D Modeling: Autodesk Fusion 360, Autodesk Meshmixer, Autodesk Mudbox, Prusaslicer, Bambu Slicer
- Visual Editing: DaVinci Resolve, Adobe Premiere Pro, Adobe Lightroom, Krita, Adobe
- Languages: English, Mandarin
- Interests: photography, drawing, graphic design, biking, Minecraft building